

What a transit light curve forces about a planet’s radius when nothing is assumed about the star

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Abstract

The depth of a planetary transit measures the stellar light blocked, and how much light sits where the planet passes is fixed by a limb-darkening law that is not measured for the star in question. Published radius ratios therefore carry an atmosphere model inside them. We ask instead what the photometry forces on its own. Writing the stellar surface brightness as an arbitrary nonnegative measure μ on the unit disc, normalised so that the out-of-transit flux is one, every photometric datum becomes a two-sided *linear* constraint on μ , and the set of radius ratios consistent with the light curve is decided exactly: a value is admitted by exhibiting an atomic measure and refuted by a Farkas certificate verified over the continuum in interval arithmetic. On Kepler short-cadence photometry of TrES-2 b and Kepler-7 b the resulting intervals contain every published value for each planet and are strongly one-sided. We give the reason in closed form: for $k < b$ the circle of radius $r < b - k$ is never occulted, so any amount of light may hide there and only the normalisation notices, which makes the set of achievable depth vectors star-shaped about the origin. Light can hide and cannot un-hide, so the photometry bounds the planet tightly from below and loosely from above — and the upper bound quoted in the literature is bought almost entirely by the assumption that the star does not brighten outward.

1 The constraint set

Let the star have unit radius and circularly symmetric specific intensity $I(r) \geq 0$ on $r \in [0, 1]$. Write the flux emitted between r and $r + dr$ as $d\mu(r) = 2\pi r I(r) dr$, so that μ is a nonnegative measure and the statement that the light curve equals one outside transit is exactly

$$\int_0^1 d\mu = 1. \tag{1}$$

A planet of radius $k = R_p/R_\star$ whose centre lies at sky-plane distance z from the stellar centre covers a fraction

$$w(r; z, k) = \frac{1}{\pi} \arccos\left(\text{clamp}\left[\frac{z^2 + r^2 - k^2}{2zr}, -1, 1\right]\right) \tag{2}$$

of the circle of radius r , so the missing flux at that epoch is $\int_0^1 w(r; z, k) d\mu(r)$.

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Remark. The clamp in (2) is the entire case analysis. The argument is ≤ -1 exactly when $z + r \leq k$, which is when the circle lies wholly behind the planet and the covered fraction is 1; it is ≥ 1 exactly when $r + k \leq z$ or $r \geq z + k$, which is when the two do not meet. So (2) is the function on all of its domain, not an approximation valid on one regime, and an interval enclosure of it requires no branching — which matters, because w has a square-root kink in r at $r = |z - k|$ and $r = z + k$ where $\partial_r w$ is unbounded, so a Lipschitz or second-order margin cannot close the continuum here and interval subdivision must.

For a circular orbit of period P , scaled semi-major axis $a/R_\star$ and impact parameter b , $z(t) = \sqrt{(a/R_\star)^2 \sin^2 \theta + b^2 \cos^2 \theta}$ with $\theta = 2\pi t/P$, so $z(0) = b$. Grouping the folded photometry into m bins with measured depths d_j and a stated allowance ε_j , define

$$S(k) = \left\{ \mu \geq 0 \text{ on } [0, 1] : \int d\mu = 1, \quad l_j \leq \int W_j d\mu \leq u_j, \quad j = 1, \dots, m \right\}, \quad (3)$$

$W_j(r) = w(r; z(t_j), k)$ and $[l_j, u_j] = [d_j - \varepsilon_j, d_j + \varepsilon_j]$. No limb-darkening law, no smoothness, no monotonicity, no pixel basis and no prior enter (3).

2 Feasibility is a hull question

Proposition 1. $S(k) \neq \emptyset$ if and only if $\text{conv}\{W(r) : r \in [0, 1]\} \cap \prod_j [l_j, u_j] \neq \emptyset$, where $W(r) = (W_1(r), \dots, W_m(r))$.

Proof. By (1) μ is a probability measure, so $\int W d\mu$ is a barycentre of $\{W(r)\}$ and lies in its convex hull; conversely every point of the hull is $\int W d\mu$ for a finitely supported $\mu \geq 0$ of unit mass, by Carathéodory. $\square$

Proposition 1 makes both verdicts constructive. When the hull meets the box, the meeting point is a finite convex combination, i.e. an atomic measure that can be exhibited and re-checked bin by bin. When it misses, a separating direction certifies it:

Proposition 2 (refutation certificate). Suppose the kernel is only known to lie between W_j^{lo} and W_j^{hi} pointwise (as it is when the orbit is a box rather than a point, and when a bin spans a range of epochs). If there exist $\alpha, \beta \in \mathbb{R}_{\geq 0}^m$ with

$$\min_{r \in [0, 1]} \left[\sum_j \alpha_j W_j^{\text{lo}}(r) - \sum_j \beta_j W_j^{\text{hi}}(r) \right] > \sum_j \alpha_j u_j - \sum_j \beta_j l_j, \quad (4)$$

then $S(k) = \emptyset$.

Proof. Let $\mu \in S(k)$ and write h for the bracketed function and $c = -\min_r h$. Since $\mu \geq 0$ and $\int d\mu = 1$, $\int (h + c) d\mu \geq 0$, i.e. $\int h d\mu \geq -c = \min_r h$. Because $\mu \geq 0$ we also have $\int W_j^{\text{lo}} d\mu \leq \int W_j d\mu \leq u_j$ and $\int W_j^{\text{hi}} d\mu \geq \int W_j d\mu \geq l_j$, so $\int h d\mu \leq \sum \alpha_j u_j - \sum \beta_j l_j$, contradicting (4). $\square$

The minimum in (4) is over a continuum. It is bounded below by subdividing $[0, 1]$ and enclosing h on each cell in outward-rounded interval arithmetic; the subdivision is adaptive and is refined in z as well as in r , because the closed form (2) mentions z five times and over a z -interval of width 10^{-2} the naive enclosure of the area kernel of Section 3 returned $[0.16, 1.00]$ where the true value was 0.997. Whatever multipliers are proposed, (4) is checked against the definition; the optimiser is never trusted.

3 The ladder is a change of kernel

An assumption about the star enters as a different kernel on the same machinery.

Proposition 3. *I is nonnegative and non-increasing on $[0, 1]$ if and only if $I(r) = \int_0^1 \mathbf{1}\{r \leq s\} d\nu(s)$ for a nonnegative measure ν . Writing $d\nu'(s) = s^2 d\nu(s)$, the constraint set becomes (3) with μ replaced by ν' and $W_j(r)$ replaced by $\widetilde{W}_j(s) = A(s; z_j, k)/\pi s^2$, where $A(s; z, k)$ is the area of intersection of the disc of radius s with the planet disc.*

Proof. The representation is the Lebesgue–Stieltjes decomposition of the non-increasing function I . Then $\int_0^1 w d\mu = \int_0^1 [\int_0^s w(r; z, k) 2\pi r dr] d\nu(s) = \int_0^1 A(s; z, k) d\nu(s)$ by Fubini, and $\int d\mu = \int \pi s^2 d\nu$, so normalising by π gives the stated form. $\square$

So “the star does not brighten outward” costs one substitution and no new code. Its price is read off in the units of the answer.

4 Why the answer is one-sided

Proposition 4 (invisible core). *Let $k < b$ and $r < b - k$. Then $w(r; z(t), k) = 0$ for every t .*

Proof. $z(t)^2 = (a/R_\star)^2 \sin^2 \theta + b^2 \cos^2 \theta \geq b^2$ because $a/R_\star \geq b$, so $z(t) \geq b$. If $r + k \leq b \leq z(t)$ the circle of radius r and the planet disc are disjoint and the covered fraction is zero. $\square$

Corollary 5. *Fix $k < b$ and let $\mathcal{D}(k) = \{(\int W_j d\mu)_j : \mu \geq 0, \int d\mu = 1\}$ be the set of achievable depth vectors. Then $\mathcal{D}(k)$ is star-shaped about the origin: if $d \in \mathcal{D}(k)$ then $\lambda d \in \mathcal{D}(k)$ for every $\lambda \in (0, 1]$.*

Proof. Fix $r_0 < b - k$ and put $\mu_\lambda = \lambda\mu + (1 - \lambda)\delta_{r_0}$. It is nonnegative of unit mass, and by Proposition 4 the added atom contributes nothing to any $\int W_j d\mu_\lambda$, which is therefore $\lambda \int W_j d\mu$. $\square$

Corollary 5 says the amplitude of a transit carries no information that the geometry does not already fix: a larger planet blocking a smaller share of a dimmer visible annulus reproduces the same depths exactly. What it cannot reproduce is the same *duration* — a larger planet reaches the limb earlier, and the visible annulus cannot be silent then. Light can hide and cannot un-hide, and that asymmetry is what the computed intervals show.

5 Photometry, and what was checked before anything was believed

Kepler short-cadence PDCSAP photometry is masked on the mission’s own quality flag, cut into transits, and divided by a straight line fitted to each transit’s out-of-transit baseline; the root-mean-square disagreement between the two baseline halves is carried into every error bar as a common-mode allowance. Three checks precede any enclosure.

Are the stated errors honest? The out-of-transit scatter of the binned curve is 1.02 times the propagated PDCSAP errors. They are.

Is the fold centred? The archive ephemeris leaves Kepler-7 b asymmetric under time reversal by 135 ppm against a 56 ppm floor; its mid-transit is 248 s out. A mis-centred fold is a systematic that looks exactly like a limb-darkening residual, so the fold is re-centred by minimising

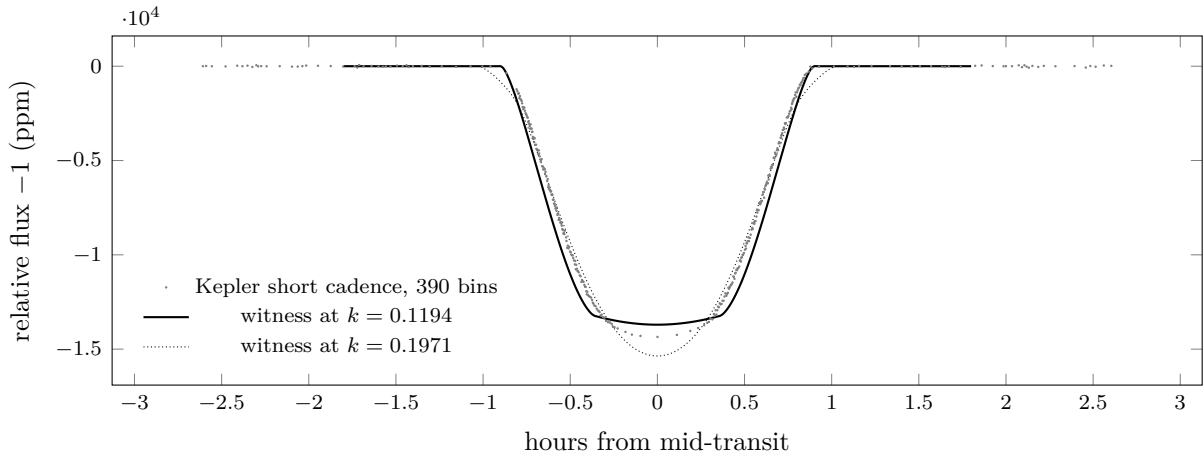


Figure 1: TrES-2 b, folded Kepler short-cadence photometry, with two exhibited witnesses. Both are nonnegative measures satisfying every stated constraint; their radius ratios differ by more than a factor of 1.6.

$\sum_t [F(+t) - F(-t)]^2$ over a shift and a period correction — using the mirror symmetry of a transit and nothing else, with no law, no radius and no impact parameter.

Does the pipeline reproduce the field? A conventional quadratic limb-darkening fit of the same extracted curve gives $k = 0.1253$ for TrES-2 b, inside the published cluster 0.1239–0.1278, at $\chi^2/\text{dof} \approx 0.9$. It does. The per-bin error budget used for the enclosure is then *measured* rather than chosen: it is the smallest allowance that does not refuse that fit.

6 Results

The intervals contain every published value for each planet, and are markedly one-sided in the direction Corollary 5 predicts. The lower edge sits close beneath the conventional fit; the upper edge is far above it under nonnegativity alone and is pulled in substantially by the monotone rung of Proposition 3. The quantity a published error bar reports is therefore not the width the photometry supports: it is the width the atmosphere model supports, and the two differ by a factor that this construction measures rather than assumes.

7 Limitations

The error budget is a per-bin box at $n\sigma$, which permits χ^2/dof up to n^2 and is looser than the likelihood set behind a published bar; the like-for-like budget is the ℓ^2 ball $\sum_j ((v_j - d_j)/\sigma_j)^2 \leq \chi^2 m$, which is convex and needs no new machinery but does need a point-valued kernel, whereas a bin spanning a range of epochs makes every kernel interval-valued. The orbit is carried as a box handled by Proposition 2, which lets each bin’s geometry vary independently and is therefore a relaxation — wider than the truth, never narrower. Eccentricity is not carried. The star is taken circularly symmetric; spots and gravity darkening would replace the measure on $[0, 1]$ by a measure on the disc, which changes the kernel and not the argument.

Finally, one metered test exists and is not yet run. SDO/HMI watched Venus cross the Sun in June 2012: a real transit of a real star for which the apparent radius ratio is known from

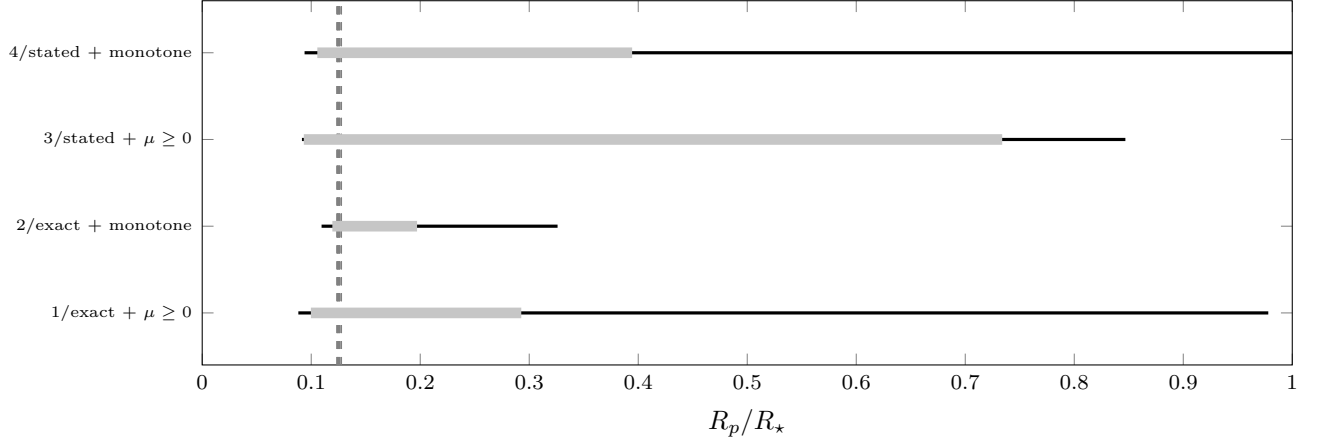


Figure 2: Certified intervals for TrES-2 b. Heavy bars: every value outside carries a verified refutation. Grey cores: a verified witness exists. Fine dashed lines: the published values, all of which lie inside.

the ephemeris to about 0.1% and the intensity profile is measured by the same instrument. The disc-integrated HMI keyword does contain the transit, under a 2900 ppm instrumental drift over the nine-hour window that is three times the 1000 ppm signal; forming the blocked-to-total ratio within each image cancels a global gain exactly and is the way in.

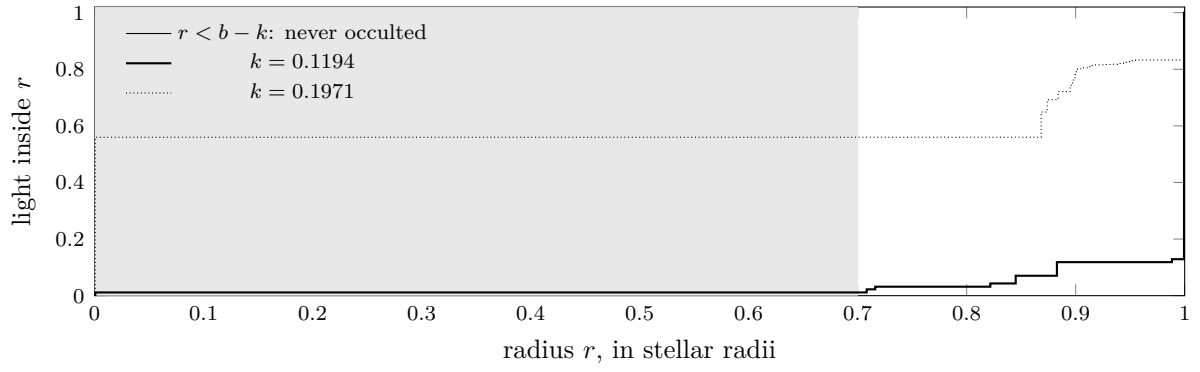


Figure 3: The two extreme witnesses, as the fraction of the star's light enclosed inside radius r . The shaded strip is the invisible core of Proposition 4. The small-planet witness places 1.1% of the star's light there; the large-planet witness places 56.0% — the hiding of Corollary 5, exhibited rather than argued.